

- endoplasmic reticulum (ER) by binding to a receptor protein on the ER.
- signal transduction** The linkage of a mechanical, chemical, or electromagnetic stimulus to a specific cellular response.
- signal transduction pathway** A series of steps linking a mechanical, chemical, or electrical stimulus to a specific cellular response.
- silent mutation** A nucleotide-pair substitution that has no observable effect on the phenotype; for example, within a gene, a mutation that results in a codon that codes for the same amino acid.
- simple fruit** A fruit derived from a single carpel or several fused carpels.
- simple sequence DNA** A DNA sequence that contains many copies of tandemly repeated short sequences.
- single bond** A single covalent bond; the sharing of a pair of valence electrons by two atoms.
- single circulation** A circulatory system consisting of a single pump and circuit, in which blood passes from the sites of gas exchange to the rest of the body before returning to the heart.
- single-lens eye** The camera-like eye found in some jellies, polychaete worms, spiders, and many molluscs.
- single nucleotide polymorphism (SNP)** (snip) A single base-pair site in a genome where nucleotide variation is found in at least 1% of the population.
- single-strand binding protein** A protein that binds to the unpaired DNA strands during DNA replication, stabilizing them and holding them apart while they serve as templates for the synthesis of complementary strands of DNA.
- sinoatrial (SA) node** (si'-nō-ā'-trē-uhl) A region in the right atrium of the heart that sets the rate and timing at which all cardiac muscle cells contract; the pacemaker.
- sister chromatids** Two copies of a duplicated chromosome attached to each other by proteins at the centromere and, sometimes, along the arms. While joined, two sister chromatids make up one chromosome. Chromatids are eventually separated during mitosis or meiosis II.
- sister taxa** Groups of organisms that share an immediate common ancestor and hence are each other's closest relatives.
- skeletal muscle** A type of striated muscle that is generally responsible for the voluntary movements of the body.
- sliding-filament model** The idea that muscle contraction is based on the movement of thin (actin) filaments along thick (myosin) filaments, shortening the sarcomere, the basic unit of muscle organization.
- slow-twitch fiber** A muscle fiber that can sustain long contractions.
- small interfering RNA (siRNA)** One of multiple small, single-stranded RNA molecules generated by cellular machinery from a long, linear, double-stranded RNA molecule. The siRNA associates with one or more proteins in a complex that can degrade or prevent translation of an mRNA with a complementary sequence.
- small intestine** The longest section of the alimentary canal, so named because of its small diameter compared with that of the large intestine; the principal site of the enzymatic hydrolysis of food macromolecules and the absorption of nutrients.
- smooth ER** That portion of the endoplasmic reticulum that is free of ribosomes.
- smooth muscle** A type of muscle lacking the striations of skeletal and cardiac muscle because of the uniform distribution of myosin filaments in the cells; responsible for involuntary body activities.
- social learning** Modification of behavior through the observation of other individuals.
- sociobiology** The study of social behavior based on evolutionary theory.
- sodium-potassium pump** A transport protein in the plasma membrane of animal cells that actively transports sodium out of the cell and potassium into the cell.
- soil horizon** A soil layer with physical characteristics that differ from those of the layers above or beneath.
- solute** (sol'-yūt) A substance that is dissolved in a solution.
- solute potential (Ψ_s)** A component of water potential that is proportional to the molarity of a solution and that measures the effect of solutes on the direction of water movement; also called osmotic potential, it can be either zero or negative.
- solution** A liquid that is a homogeneous mixture of two or more substances.
- solvent** The dissolving agent of a solution. Water is the most versatile solvent known.
- somatic cell** (sō-mat'-ik) Any cell in a multicellular organism except a sperm or egg or their precursors.
- somite** One of a series of blocks of mesoderm that exist in pairs just lateral to the notochord in a vertebrate embryo.
- soredium** (suh-rē'-dē-um) (plural, **soredia**) In lichens, a small cluster of fungal hyphae with embedded algae.
- sorus** (plural, **sori**) A cluster of sporangia on a fern sporophyll. Sori may be arranged in various patterns, such as parallel lines or dots, which are useful in fern identification.
- spatial learning** The establishment of a memory that reflects the environment's spatial structure.
- speciation** (spē'-sē-ā'-shun) An evolutionary process in which one species splits into two or more species.
- species** (spē'-sēz) A population or group of populations whose members have the potential to interbreed in nature and produce viable, fertile offspring but do not produce viable, fertile offspring with members of other such groups.
- species-area curve** (spē'-sēz) The biodiversity pattern that shows that the larger the geographic area of a community is, the more species it has.
- species diversity** (spē'-sēz) The number and relative abundance of species in a biological community.
- species richness** (spē'-sēz) The number of species in a biological community.
- specific heat** The amount of heat that must be absorbed or lost for 1 g of a substance to change its temperature by 1°C.
- spectrophotometer** An instrument that measures the proportions of light of different wavelengths absorbed and transmitted by a pigment solution.
- sperm** The male gamete.
- spermatheca** (sper'-muh-thē'-kuh) (plural, **spermathecae**) In many insects, a sac in the female reproductive system where sperm are stored.
- spermatogenesis** (sper-ma'-tō-gen'-uh-sis) The continuous and prolific production of mature sperm in the testis.
- spermatogonium** (sper-ma'-tō-gō'-nē-um) (plural, **spermatogonia**) A cell that divides mitotically to form spermatocytes.
- S phase** The synthesis phase of the cell cycle; the portion of interphase during which DNA is replicated.
- sphincter** (sfink'-ter) A ringlike band of muscle fibers that controls the size of an opening in the body, such as the passage between the esophagus and the stomach.
- spiral cleavage** A type of embryonic development in protostomes in which the planes of cell division that transform the zygote into a ball of cells are diagonal to the vertical axis of the embryo. As a result, the cells of each tier sit in the grooves between cells of adjacent tiers.
- spliceosome** (spli'-sō-sōm) A large complex made up of proteins and RNA molecules that splices RNA by interacting with the ends of an RNA intron, releasing the intron and joining the two adjacent exons.
- spongocoel** (spon'-jō-sēl) The central cavity of a sponge.
- spontaneous process** A process that occurs without an overall input of energy; a process that is energetically favorable.
- sporangium** (spōr-an'-jē-um) (plural, **sporangia**) A multicellular organ in fungi and plants in which meiosis occurs and haploid cells develop.
- spore** (1) In the life cycle of a plant or alga undergoing alternation of generations, a haploid cell produced in the sporophyte by meiosis. A spore can divide by mitosis to develop into a multicellular haploid individual, the gametophyte, without fusing with another cell. (2) In fungi, a haploid cell, produced either sexually or asexually, that produces a mycelium after germination.
- sporocyte** (spō'-ruh-sit) A diploid cell within a sporangium that undergoes meiosis and generates haploid spores; also called a spore mother cell.
- sporophyll** (spō'-ruh-fil) A modified leaf that bears sporangia and hence is specialized for reproduction.
- sporophyte** (spō'-ruh-fit) In organisms (plants and some algae) that have alternation of generations, the multicellular diploid form that results from the union of gametes. Meiosis in the sporophyte produces haploid spores that develop into gametophytes.
- sporopollenin** (spōr-uh-pol'-eh-nin) A durable polymer that covers exposed zygotes of charophyte algae and forms the walls of plant spores, preventing them from drying out.
- stability** In evolutionary biology, a term referring to a hybrid zone in which hybrids continue to be produced; this causes the hybrid zone to be "stable" in the sense of persisting over time.